THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 3

Chapter 9: **The Amazing World of Solutes, Solvents and Solutions** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The solubility of gases in water: 1

(a) decreases with an increase in temperature
(b) is not affected
(c) is infinite
(d) increases with temperature

2. The solubility of most solids in water: 1

(a) increases with temperature **(b)** decreases with temperature
(c) is zero **(d)** is unaffected by temperature

3. To make a dilute solution from a concentrated one, we: 1

(a) add more solute **(b)** add more solvent
(c) heat it **(d)** evaporate the solvent

4. A cold drink fizzes more when warm because: 1

(a) gas dissolves more **(b)** sugar dissolves more
(c) water boils **(d)** dissolved carbon dioxide escapes more

5. Which of the following does NOT dissolve in water? 1

(a) sand **(b)** sugar **(c)** salt **(d)** alcohol

6. Concentrated solution has: 1

(a) no solvent
(b) no solute
(c) a large amount of solute in a given volume
(d) very little solute

7. Stirring a solution: 1

(a) stops dissolving **(b)** speeds up dissolving
(c) removes solute **(d)** slows dissolving

8. Which is a gas dissolved in a liquid? 1

(a) brass **(b)** sugar solution **(c)** soda water **(d)** salt solution

9. The particles of a true solution: 1

(a) settle down on standing
(b) can be seen with a hand lens
(c) can be seen with the naked eye
(d) cannot be seen even with an ordinary microscope

10. Crushing sugar crystals before dissolving them: 1

(a) increases the rate of dissolving **(b)** decreases the rate
(c) changes the solvent **(d)** stops dissolving

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Give two examples of gaseous solutions. 2

12. Why is water called the universal solvent? 2

13. Name two factors that increase the rate of dissolving. 2

14. Why does a soft drink lose its fizz when left open? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Differentiate between a solution and a colloid in three points. 3

16. Why do we stir sugar in tea? Explain the effect of stirring. 3

17. How can an unsaturated solution be made saturated? Give three ways. 3

OR

Find the mass percentage of a solution with 30 g sugar dissolved in 120 g water. 3

18. Why does sugar powder dissolve faster than sugar crystals? 3

OR

Differentiate between a pure solvent and a solution using conductivity of distilled and tap water. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain the factors that affect the rate of dissolving, designing a fair test for the effect of temperature. 5

OR

Explain how crystals of sugar or copper sulphate can be prepared from a hot saturated solution. 5

20. Explain why water is an important solvent for life, with five examples. 5

OR

Discuss why soft drinks are stored under pressure and kept cool. 5

END OF QUESTION PAPER — Total: 40 Marks